

DSA 3050A — Business Intelligence & Visualization

Project Report

Business problem, data preparation, modelling, DAX, dashboard design, insights and recommendations

Document	Full Project Report
Group	Grp05
Topic	Option 4 — Airline Delay and Operations Dashboard
Dataset	hflights — 227,496 Houston departures, 2011
Source	US DOT Bureau of Transportation Statistics
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1. Executive summary

This project analyses 227,496 commercial flights that departed Houston's two airports during 2011, and delivers a Power BI dashboard that attributes delay and cancellation to the carrier, route, airport, hour and month that produced them.

The headline position is poor: 47.8% of flights that operated arrived late, average arrival delay was 7.1 minutes, and 2,973 flights were cancelled outright. But the analysis shows that this headline is made of four separable problems, not one general one. Three carriers fly 82% of the network and all three run below average. Delay is created hour-by-hour through the operating day rather than inherited from the schedule. Cancellations are dominated by a single February ice storm which supplies 37% of the annual total. And the metrics currently in use — average delay, blended cancellation rate — point management at the least actionable parts of all three.

Six recommendations follow, of which the cheapest — changing what is measured — unlocks the value of the other five.

2. Business problem and data understanding

2.1 The business problem

Flight delays are the largest controllable cost in airport and airline operations. A delayed departure burns fuel while taxiing, pushes crews towards duty-hour limits, forces gate reallocation, breaks passenger connections and, past a threshold, triggers compensation. A cancellation destroys the revenue of the whole rotation and displaces passengers onto later services that are already full.

Houston handled 227,496 departures in 2011 and 47.8% of operated flights arrived late. Management could see that the network was late but not where the lateness was being created — whether it sat with particular carriers, routes, times of day or seasons. Without that attribution, mitigation spending is guesswork. This project supplies the attribution.

2.2 Target organization

The dashboard is written for the Operations Performance team of the Houston Airport System, the municipal authority that runs both IAH and HOU, together with the network operations centres of its two dominant carriers. It serves three decisions: where the COO should direct capital and staffing; which routes and departure banks network planners should re-time; and which carriers and periods duty managers should cover.

2.3 The dataset

Property	Value
Dataset	hflights — all commercial flights departing Houston, 2011
Source	US DOT Bureau of Transportation Statistics, via the R hflights package
URL	https://cran.r-project.org/package=hflights
Licence	Public domain (US federal work), redistributed under GPL-2
Rows	227,496
Columns	21 raw, 32 after cleaning
Period	1 January – 31 December 2011
Origin airports	2 — IAH (175,197), HOU (52,299)
Destinations	116
Airlines	15
Routes	149

Property	Value
Aircraft	3,319 distinct tail numbers
Cancelled	2,973 (1.31%)
Diverted	649 (0.29%)

The data is genuine government reporting. Nothing in this project is synthetic, generated or manually invented.

2.4 Key business questions

- Which airlines have the highest delays, and does poor punctuality concentrate in the carriers that fly the most?
- Which routes are the most delayed, and are they delayed enough to justify re-timing?
- How do the two Houston airports compare on departure delay, arrival delay and taxi-out?
- What causes cancellations, how are those causes distributed, and how much is avoidable?
- How does delay trend by month and season, and when is the network most fragile?
- What are the peak travel periods, and does traffic volume by itself explain delay?
- Does the time of day a flight is scheduled predict how late it will be?
- How much departure delay do carriers recover in the air before arrival?

3. Power Query data cleaning and transformation

Each step below is an Applied Step in the Power Query Editor, and the exported cleaned CSV in Dataset/ can be checked against the model at any time. Row count is identical before and after: no flights were dropped, because cancelled and diverted flights are the subject of part of the analysis.

3.1 Profiling first

Column profiling was enabled across the full 227,496 rows rather than the 1,000-row default sample. This is what set the cleaning plan. It showed a source that was structurally clean — zero duplicate rows, zero blank rows, consistent codes — but with three properties that would have produced wrong answers if handled naively: times stored as HHMM integers, flags stored as 0/1 numerics, and a 98.69% null rate on CancellationCode.

Column profile of the raw dataset (all 21 columns)

Computed from Dataset/raw_dataset.csv. This is the profile that drove the cleaning plan: CancellationCode is 98.69% null by design, and the delay/time columns carry the cancelled and diverted flights as nulls.

Column	Detected type	Null count	Null %	Distinct values	Value range
Year	int64	0	0.00%	1	2011 to 2011
Month	int64	0	0.00%	12	1 to 12
DayofMonth	int64	0	0.00%	31	1 to 31
DayOfWeek	int64	0	0.00%	7	1 to 7
DepTime	float64	2,905	1.28%	1,207	1 to 2400
ArrTime	float64	3,066	1.35%	1,283	1 to 2400
UniqueCarrier	object	0	0.00%	15	AA to YV
FlightNum	int64	0	0.00%	3,740	1 to 7290
TailNum	object	795	0.35%	3,319	N0EGMQ to N9EAMQ
ActualElapsedTime	float64	3,622	1.59%	435	34 to 575
AirTime	float64	3,622	1.59%	398	11 to 549
ArrDelay	float64	3,622	1.59%	463	-70 to 978
DepDelay	float64	2,905	1.28%	429	-33 to 981
Origin	object	0	0.00%	2	HOU to IAH
Dest	object	0	0.00%	116	ABQ to XNA
Distance	int64	0	0.00%	159	79 to 3904
TaxiIn	float64	3,066	1.35%	96	1 to 165
TaxiOut	float64	2,947	1.30%	143	1 to 163
Cancelled	int64	0	0.00%	2	0 to 1
CancellationCode	object	224,523	98.69%	4	A to D
Diverted	int64	0	0.00%	2	0 to 1

Column profile of the raw dataset, computed over all 227,496 rows.

3.2 The ten required cleaning tasks

#	Task	What was done
1	Rename unclear columns	Thirteen columns renamed to business language: DayofMonth→Day, UniqueCarrier→Airline Code, TailNum→Aircraft ID, ArrDelay→Arrival Delay, DepDelay→Departure Delay, and so on.
2	Correct data types	Delay, taxi and elapsed-time columns forced from decimal to whole number; Flight Date built as a true Date; codes set to Text; flags converted from 0/1 numerics to Yes/No text.
3	Remove duplicates and blank rows	Table.Distinct across all columns. The source proved clean — 0 rows removed. This was verified rather than assumed.
4	Trim and clean text	Text.Trim and Text.Upper applied to the five text code columns to eliminate padding and case drift.
5	Replace inconsistent values	Cancellation codes A/B/C/D replaced with Carrier / Weather / National Air System / Security; Cancelled and Diverted flags replaced with No/Yes.
6	Handle missing values	795 null Aircraft IDs replaced with "Unknown". The 224,523 null cancellation codes are null by design and were mapped to "Not Cancelled". The 3,622 null Arrival Delays were deliberately left null — see 3.4.
7	Remove unnecessary columns	Year (constant 2011) dropped, along with the raw DepTime, ArrTime, Cancelled, Diverted and CancellationCode columns once decoded replacements existed.
8	Split and merge columns	DepTime/ArrTime split from HHMM integers into HH:MM clock text plus a numeric Departure Hour; Origin and Destination merged into a single Route key.

#	Task	What was done
9	Custom and conditional columns	Flight Status, Delay Category, Distance Band, Time of Day, Season and Delay Recovered created as nested conditional columns.
10	Extract date parts	Month, Month Name, Quarter, Day and Day of Week derived from Flight Date.

3.3 Advanced Power Query tasks — 8 completed against a requirement of 6

Task	Where it was applied
Merge queries on a common key	Dim_Airline merged into the fact table on Airline Code to attach readable carrier names.
Create a date table	Dim_Date generated across 2011 with Year, Month, Month Name, Quarter, Day, Day of Week, Week of Year and Is Weekend.
Group By with multiple aggregations	Summary_Airline_Performance built by grouping on Airline and aggregating Total Flights, Average Arrival Delay and Total Cancellations in one step.
Create summarized tables	The same table — a 15-row aggregate that keeps the ranking and ribbon visuals off the 227,496-row fact table.
Business categories with nested conditions	Delay Category (5 bands), Flight Status (4 states), Distance Band (4 bands) and Time of Day (4 bands).
Create reference queries	Dim_Airline and Summary_Airline_Performance are reference queries off the cleaned fact query, so upstream changes flow to both.
Use column profiling	Enabled over the whole table rather than the default sample — this surfaced the 98.69% null rate on CancellationCode.
Remove or handle errors	try...otherwise wrapped around the HHMM conversion so the 2400 values present in the source roll to 00:00 instead of throwing.

3.4 The most consequential cleaning decision

The 3,622 null values in Arrival Delay belong to cancelled and diverted flights. The obvious move — replace nulls with zero — would have been wrong in a way that is easy to miss: a zero reads as "arrived exactly on time", which would simultaneously pull every delay average towards zero and inflate the on-time rate by counting 3,622 flights that never arrived as punctual. They were left null, and every rate measure in the model divides by completed flights instead of by total flights. A cancelled flight is neither on time nor delayed; it is a different kind of failure and is counted separately.

Cleaned dataset — 227,496 rows x 32 columns after Power Query transformation

Rendered directly from Dataset/cleaned_dataset.csv (first 14 rows, 16 of 32 columns shown). Codes are decoded to names, times are real clock values, and the derived business categories are in place.

Flight Date	Month Name	Quarter	Season	Day of Week	Airline	Aircraft ID	Route	Distance Band	Departure Time	Time of Day	Departure Delay	Arrival Delay	Delay Category	Flight Status	Cancellation Reason
2011-01-01	Jan	Q1	Winter	Sat	American Airlines	N576AA	IAH-DFW	Short Haul	14:00	Afternoon	0	-10	On Time / Early	On Time	Not Cancelled
2011-01-02	Jan	Q1	Winter	Sun	American Airlines	N557AA	IAH-DFW	Short Haul	14:01	Afternoon	1	-9	On Time / Early	On Time	Not Cancelled
2011-01-03	Jan	Q1	Winter	Mon	American Airlines	N541AA	IAH-DFW	Short Haul	13:52	Afternoon	-8	-8	On Time / Early	On Time	Not Cancelled
2011-01-04	Jan	Q1	Winter	Tue	American Airlines	N403AA	IAH-DFW	Short Haul	14:03	Afternoon	3	3	Minor Delay	Delayed	Not Cancelled
2011-01-05	Jan	Q1	Winter	Wed	American Airlines	N492AA	IAH-DFW	Short Haul	14:05	Afternoon	5	-3	On Time / Early	On Time	Not Cancelled
2011-01-06	Jan	Q1	Winter	Thu	American Airlines	N262AA	IAH-DFW	Short Haul	13:59	Afternoon	-1	-7	On Time / Early	On Time	Not Cancelled
2011-01-07	Jan	Q1	Winter	Fri	American Airlines	N493AA	IAH-DFW	Short Haul	13:59	Afternoon	-1	-1	On Time / Early	On Time	Not Cancelled
2011-01-08	Jan	Q1	Winter	Sat	American Airlines	N477AA	IAH-DFW	Short Haul	13:55	Afternoon	-5	-16	On Time / Early	On Time	Not Cancelled
2011-01-09	Jan	Q1	Winter	Sun	American Airlines	N476AA	IAH-DFW	Short Haul	14:43	Afternoon	43	44	Moderate Delay	Delayed	Not Cancelled
2011-01-10	Jan	Q1	Winter	Mon	American Airlines	N504AA	IAH-DFW	Short Haul	14:43	Afternoon	43	43	Moderate Delay	Delayed	Not Cancelled
2011-01-11	Jan	Q1	Winter	Tue	American Airlines	N655AA	IAH-DFW	Short Haul	14:29	Afternoon	29	29	Moderate Delay	Delayed	Not Cancelled
2011-01-12	Jan	Q1	Winter	Wed	American Airlines	N577AA	IAH-DFW	Short Haul	14:19	Afternoon	19	5	Minor Delay	Delayed	Not Cancelled
2011-01-13	Jan	Q1	Winter	Thu	American Airlines	N476AA	IAH-DFW	Short Haul	13:58	Afternoon	-2	-9	On Time / Early	On Time	Not Cancelled
2011-01-14	Jan	Q1	Winter	Fri	American Airlines	N552AA	IAH-DFW	Short Haul	13:57	Afternoon	-3	-6	On Time / Early	On Time	Not Cancelled

The cleaned table: codes decoded, times as real clock values, business categories in place.

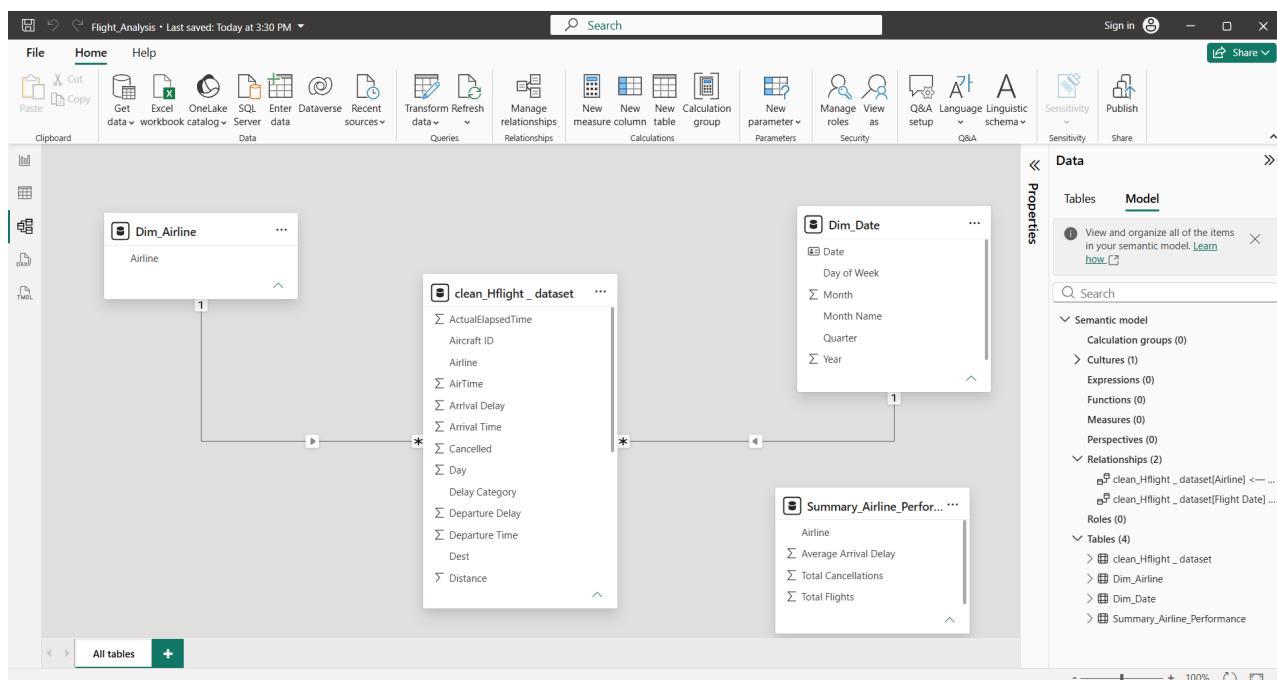
4. Data modelling

The model is a star schema: one fact table, three dimensions, and a disconnected table holding the measures. The raw extract arrived as a single wide denormalised table; carrier and calendar attributes were lifted out into dimensions so that slicers filter through a small dimension rather than scanning 227,496 rows of repeated text.

Table	Role	Grain and contents
Fact_Flights	Fact	One row per flight — 227,496 rows, 32 columns.
Dim_Date	Dimension	One row per calendar day of 2011 (365 rows). Marked as the model date table.
Dim_Airline	Dimension	One row per carrier (15 rows).
Summary_Airline_Performance	Aggregate	One row per carrier with pre-computed totals.
_Measures	Disconnected	Holds all DAX measures; contains no data.

4.1 Relationships

From	To	Cardinality	Direction
Fact_Flights[Flight Date]	Dim_Date[Date]	Many-to-one	Single
Fact_Flights[Airline]	Dim_Airline[Airline]	Many-to-one	Single
Summary_Airline_Performance[Airline]	Dim_Airline[Airline]	Many-to-one	Single



The star schema as built in Power BI.

4.2 Design decisions

No flat table. Dimensions were extracted rather than left inline.

Single-direction filters throughout. No bi-directional relationships, so there is no ambiguity in the filter path and no risk of a circular dependency.

A real date table. Dim_Date is a contiguous 365-day calendar flagged with Mark as date table — which is what makes TOTALYTD, DATESQTD, DATESMTD and PREVIOUSMONTH valid rather than silently wrong.

Measures isolated. All measures live in _Measures, a table with no relationships and no columns, so the field list stays readable.

An aggregate table where it pays. Summary_Airline_Performance serves the ranking and ribbon visuals from 15 rows instead of 227,496.

5. DAX measures

22 named measures sit in six display folders inside _Measures, plus one helper measure used as the shared denominator for every rate.

Folder	Measures	Count
Basic Measures	Total Flights, Total Airlines, Total Routes, Average Arrival Delay, Average Departure Delay	5
Delay Performance	Delayed Flights, On-Time Flights, Severely Delayed Flights, On-Time Rate, Delayed Flight Rate, Severe Delay Rate, Average Flights per Route	7
Time Intelligence	Flights MTD, Flights QTD, Flights YTD, Previous Month Flights	4
Rankings	Airline Delay Rank, Airline Flight Rank	2
KPI Status	On-Time Performance Status, Arrival Delay Status	2
Dynamic Titles	Airline Performance Title, Delay Trend Title	2

5.1 The denominator decision

Every rate measure divides by completed flights, not by all flights:

```
Completed Flights =
CALCULATE ( COUNTROWS ( Fact_Flights ), NOT ISBLANK ( Fact_Flights[Arrival Delay] ) )
```

```
On-Time Rate = DIVIDE ( [On-Time Flights], [Completed Flights] )
```

A cancelled flight is neither on time nor delayed. Including the 3,622 cancelled and diverted flights in the denominator would understate punctuality by roughly 1.6 percentage points and would make the on-time rate move whenever cancellations moved, which is a different phenomenon.

5.2 Representative measures

```
Severe Delay Rate =
DIVIDE ( [Severely Delayed Flights], [Completed Flights] )
```

```
Airline Delay Rank =
RANKX ( ALL ( Dim_Airline[Airline] ), [Average Arrival Delay], , DESC, DENSE )
```

```
On-Time Performance Status =
SWITCH ( TRUE (),
    [On-Time Rate] >= 0.80, "Good",
    [On-Time Rate] >= 0.65, "Watch",
    "Needs Attention" )
```

```
Airline Performance Title =
"Airline Performance for " & SELECTEDVALUE ( Dim_Airline[Airline], "All Airlines" )
```

```
Flights YTD = TOTALYTD ( [Total Flights], Dim_Date[Date] )
```

5.3 Formatting conventions

Rate measures are formatted as percentage to one decimal place; delay measures as whole numbers with a " min" suffix; counts with a thousands separator and no decimals. Measure names use Title Case and never repeat their table name.

6. Dashboard design and visuals

The report uses the Corporate Blue theme recommended for airline projects — a white/light-grey canvas, navy blue accent, sky blue and teal secondaries, and orange/red reserved strictly for warning states so that a colour never carries meaning casually.

Page	Purpose	Visuals
1 — Executive Summary	The one-screen answer to "how did Houston perform in 2011"	4 KPI cards (Total Flights, On-Time Rate, Total Routes, Average Arrival Delay) · ribbon chart of airline volume by month · decomposition tree · delay-category pie chart · navigation buttons
2 — Trend Analysis	How performance moved through the year	Monthly delay line chart · airline, month and delay-category slicers, synced across pages · dynamic title driven by Delay Trend Title
3 — Geographic & Route Analysis	Where in the network the delay sits	Azure map of flight volume by destination, bubble-sized by traffic · airline performance matrix with conditional formatting (a delay heat map) · airline and destination slicers
4 — Route Drill-through	Flight-level detail for one selected route	Not yet built — see section 8

6.1 Design rules applied

Consistent palette across all pages, with warning colours reserved for status states only.

Dynamic titles on the carrier and trend visuals, so a heading always names the current selection rather than a generic label.

Conditional formatting on the performance matrix, so the matrix doubles as the delay heat map.

Slicers synced across pages, so a carrier selected on one page stays selected on the next.

Navigation buttons on every page rather than relying on the page tabs.

Thousands separators and explicit units on every number, so no figure is ambiguous.

7. Insights and recommendations

Stated in full, with supporting charts, in Insights.pdf. Summarised here.

7.1 Insights

#	Insight	The evidence
1	Punctuality is inversely related to scale — the carriers that fly Houston most are the ones that run it late	ExpressJet 50.6%, Continental 50.9%, Southwest 53.6% on 82% of flights, against a 52.2% average and 69.7% for the best carrier.
2	Delay is manufactured during the day, not inherited from the schedule	06:00 departures average -0.0 min delay; 20:00 departures average 25.5 min.
3	Cancellations are not a chronic problem — they are one week in February	1,108 of 2,973 cancellations in February; 929 of them weather. Ex-February rate 0.89% against 1.31% blended.
4	Carrier-controllable causes dominate every month except the storm	Carrier is the top reason in 8 of 11 non-February months: 1,032 against 723 weather.
5	IAH is operationally slower on the ground but still arrives earlier than HOU	IAH taxi-out 16.9 min against HOU 9.1, yet IAH departure delay 8.4 against 12.8.
6	Delay peaks in late spring, not in summer or at Christmas	May 13.1 min against November 3.2 min. July busiest but only 4th worst. $r = 0.33$ between volume and delay.
7	Most delay is small, but a thin tail does the damage	79% of flights under 15 min late; 4.7% (10,584) an hour or more; 1,110 over three hours.
8	Two-thirds of departure delay is recovered in the air	64.9% of late departures arrive less late; 2.3 min recovered on average.

7.2 Recommendations

#	Recommendation	Expected impact
1	Target the three high-volume carriers, not the fifteen-carrier average	Moving those three by nine points moves the entire Houston system by roughly seven. Moving all twelve remaining carriers to perfection would move it by under three. This is the single highest-leverage intervention available.
2	Protect the morning bank and re-time the evening bank	Attacks delay at the point of creation rather than absorbing it downstream, and stops one late morning aircraft from producing four late evening flights.
3	Fix HOU's turnaround, not its taxiways	Redirects capital away from an airfield that is not the constraint and towards the gate process that demonstrably is.
4	Report cancellations with February separated out	The blended 1.31% figure hides a well-run operation behind one ice storm, and drives the wrong conclusion — that weather is the dominant cause, when carrier-controllable cancellations are more numerous in 8 of 11 normal months.
5	Change the operational target from average delay to severe-delay rate	Aims resource at the 10,584 flights that break connections instead of at the 1-to-14-minute band, which is largely noise and is already absorbed by schedule padding.
6	Load-plan for May, not for July	Aligns contingency spend with when the network actually fails rather than with when it is busiest — the two are not the same month.

8. Limitations

Stated plainly, because they bound what the recommendations can claim.

Single year, single origin city. The data covers 2011 departures from Houston only. Seasonal conclusions rest on one observation of each month, so May 2011 being the worst month is a fact about 2011, not a proven annual pattern. A multi-year extract would be needed to distinguish season from year.

No delay-cause breakdown. The source records cancellation reasons but not delay reasons — there is no carrier/weather/NAS/security split on the delay minutes themselves. Attribution of delay to cause is therefore inferred from timing and pattern, not read directly.

Departures only. Arrivals into Houston are not in the dataset, so inbound aircraft cannot be traced. The rotation-compounding argument in Insight 2 is inferred from the hour-of-day gradient rather than followed tail-by-tail through the day.

No passenger or cost data. Load factors, fare data and compensation costs are absent, so the financial impact of the recommendations is argued in operational terms rather than quantified in dollars.

Age of the data. 2011 predates several fleet and schedule changes, including the Continental–United merger integration. The method transfers; the specific carrier findings are historical.

One page outstanding. The route drill-through page is not yet built, so flight-level interrogation of a single route is currently done through slicers rather than through drill-through.

9. Conclusion

Houston in 2011 does not have a general punctuality problem. It has three addressable problems and one unavoidable one, and its current metrics obscure all four. 47.8% of flights arriving late is a real number, but it is produced almost entirely by three carriers, created hour-by-hour through the operating day, and reported through an average that is dominated by delays too small to matter.

The dashboard makes each of those separable at a glance, which is what turns a year of flight records into an operational decision tool. The single cheapest recommendation — report severe-delay rate instead of average delay, and separate February from the cancellation baseline — costs nothing and changes what management is aiming at.